

Units

- 2.1 Role of Data in AI & ML
- 2.2 Types of Data
- 2.3 Data Collection
- 2.4 Exploratory Data Analysis
- 2.5 Data Preprocessing

OVERVIEW : MODULE 2

Data Fundamentals

Learning Outcomes

- Gain a comprehensive understanding of the importance of data in AI and ML
- Learn about the different types of data, including structured, unstructured, and semi-structured data
- Explore various techniques for data collection, preprocessing, cleaning, and exploratory data analysis
- Learn the concepts of Exploratory Data Analysis (EDA)

2.1

Role of Data in AI & ML

Lesson Learning Outcomes

- ✓ Understand the foundational importance of data in AI and ML systems
- ✓ Recognize the significance of data quality, quantity, diversity, and timeliness in fueling accurate model training and decision-making
- ✓ Identify the challenges of data management, including issues related to data quantity, quality, and security
- ✓ Explore considerations such as ethical data collection practices, data cleaning, and data governance in ensuring reliable and responsible AI development

LESSON OVERVIEW

Module 2 Data Fundamentals

Lesson 2.1 Role of Data in AI & ML

In this lesson, we'll explore the critical role of data in driving **Artificial Intelligence (AI)** and **Machine Learning (ML)** systems. Just as fuel powers a car, data fuels intelligent systems, enabling accurate model training and decision-making.

We'll delve into the importance of data quality, quantity, diversity, and timeliness, along with the challenges and considerations associated with data management in AI and ML.

- **Foundation:** Data is the bedrock of AI and ML systems
- **Fuel:** High-quality data fuels accurate model training and predictions
- **Decision-Making:** Data-driven decisions lead to better outcomes
- **Exploration:** We'll delve into data quality, quantity, diversity, and timeliness
- **Discussion:** Address challenges and considerations in data management for AI and ML

Why Data Matters

The Fuel of AI & ML

Foundation

Data is the bedrock of artificial intelligence (AI) and machine learning (ML) systems.

Fuel

High-quality data fuels accurate model training and predictions.

Decision-Making

Data-driven decisions lead to better outcomes.

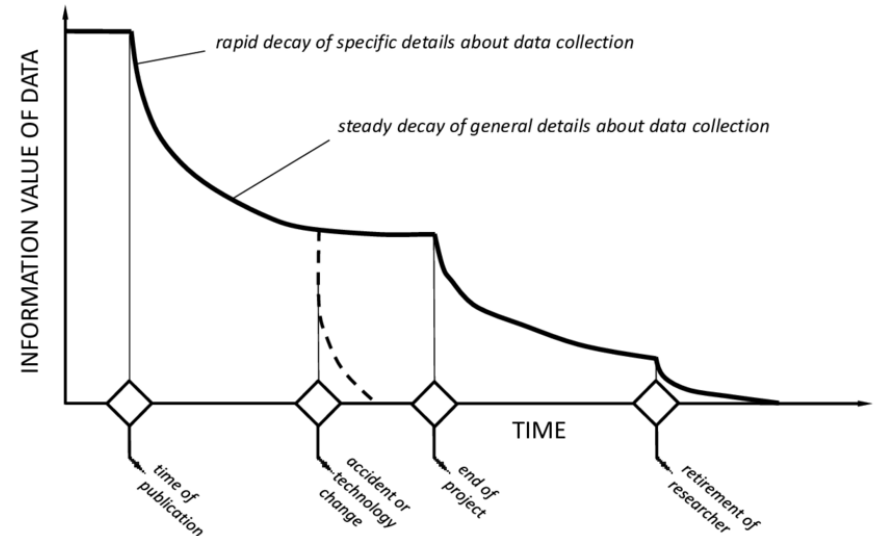
Imagine a car without fuel. That's what AI and Machine Learning (ML) are like without data. Data serves as the fuel that powers these intelligent systems. Just like a car's performance depends on the quality and quantity of fuel, the effectiveness of AI models hinges on the quality and volume of data they are trained on. The more data an AI model has access to, the better it can learn and improve its ability to perform tasks. In the following slides, we'll explore why data is so crucial in AI and ML, the different types of data used, and the challenges associated with data management.

The Role of Data

Aspect	Importance
Quality	Clean, relevant data ensures robust models
Quantity	Sufficient data prevents overfitting and improves generalization
Diversity	Diverse data captures real-world variations and avoid bias
Timeliness	Fresh data adapts models to changing environments

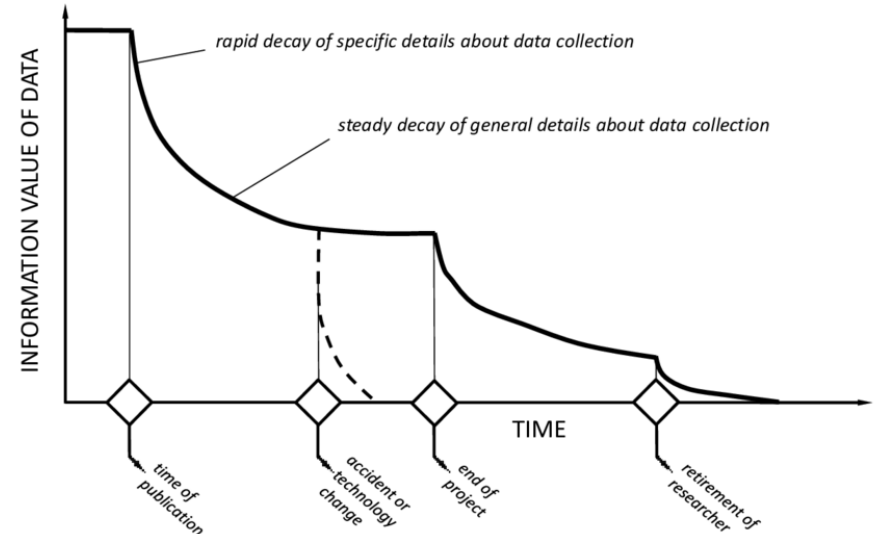
Time Value of Data

- **Time value of data** refers to the notion that the relevance and usefulness of data diminish over time
- In many applications, such as financial trading or weather forecasting, recent data **holds more value** than older data
- **Timeliness of data** is crucial for making informed decisions and predictions, especially in dynamic environments
- **Delayed or outdated data** may lead to inaccurate analyses or decisions, potentially resulting in missed opportunities or suboptimal outcomes



Time Value of Data

- **Real-time data** processing and analysis techniques are employed to leverage the time value of data, enabling swift decision-making and response to changing conditions
- **Proper data management strategies**, including data retention policies and data streaming technologies, help organizations maximize the value of their data over time



Data Challenges and Considerations

Always Keep an Eye on These

Challenges of Data Management

- **Data Quantity:** Having enough data is crucial, but managing massive datasets can be challenging.
- **Data Quality:** Inaccurate or incomplete data can lead to biased and unreliable AI models.
- **Data Security:** Protecting sensitive data and ensuring responsible AI development is essential.

Considerations

- **Data Collection:** Ethical data collection practices and user privacy must be prioritized.
- **Data Cleaning:** Ensuring data accuracy and completeness is crucial for reliable AI models.
- **Data Governance:** Establishing clear guidelines for data ownership, usage, and security is essential.

Role of Data in AI & ML

SUMMARY

- ✓ Data is fundamental to AI and ML, driving intelligence and decision-making.
- ✓ Quality, quantity, diversity, and timeliness of data are crucial for accurate predictions.
- ✓ Challenges include managing data quantity, quality, and security.
- ✓ Considerations involve ethical data collection practices and clear governance guidelines.

2.2

Types of Data

Lesson Learning Outcomes

- ✓ Recognize the distinctions between structured, unstructured, and semi-structured data
- ✓ Understand the characteristics and examples of each type of data
- ✓ Appreciate the importance of structured data for efficient storage, retrieval, and analysis
- ✓ Explore the challenges and opportunities associated with unstructured data

LESSON OVERVIEW

Module 2 Data Fundamentals

Lesson 2.2 Types of Data

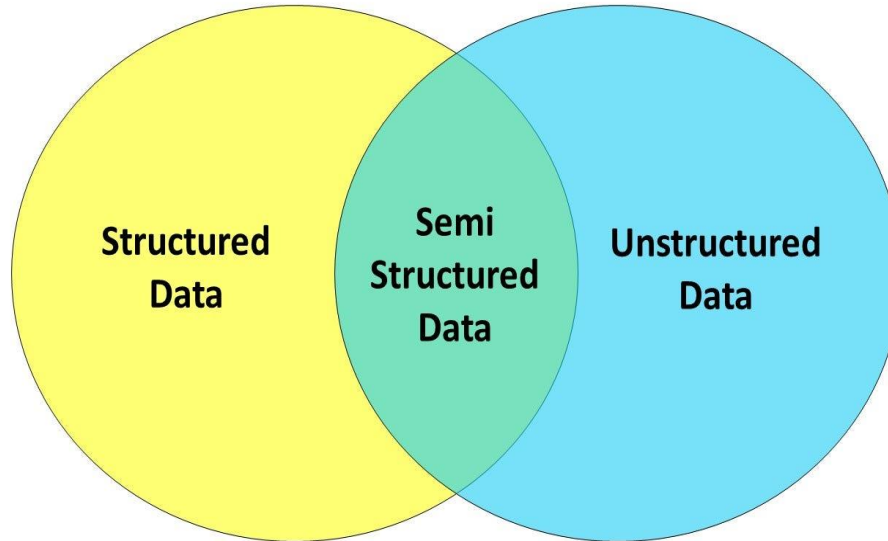
This lesson introduces the fundamental types of data crucial for **Artificial Intelligence (AI)** and **Machine Learning (ML)**. We'll cover structured data, which is organized into tables, making it easy to query and analyze. Then, we'll explore unstructured data, lacking a predefined structure but holding valuable insights requiring advanced techniques like **NLP**.

Lastly, we'll discuss semi-structured data, offering flexibility with a mix of structured and unstructured elements. Understanding these data types is essential for effective data management and analysis in AI and ML.

- Definitions and examples of different types of data
- Characteristics of different types of data
- Importance of structured data
- Challenges and opportunities of unstructured data

Types of Data

Three Basic Types



Structured Data

Data that follows a strict format and is organized into tables with clearly defined **relationships**.

Examples

- Relational databases
- Spreadsheets
- CSV files

Characteristics

Easy to query, analyze, and process using traditional Database Management Systems

UNSTRUCTURED DATA



VS

STRUCTURED DATA



Unstructured Data

Most of The Data Produced are Unstructured

Data that lacks a predefined data model or structure, making it challenging to organize and analyze.

Examples

- Text documents
- Images
- Audio files and video files

Characteristics

Contains rich information but requires advanced techniques such as natural language processing (NLP) and computer vision for analysis.

Semi-Structured Data

Mix of Two Types

Data that does not conform to a rigid structure but contains some organizational properties.

Examples

- XML files
- JSON files
- NoSQL databases.

Characteristics

Offers flexibility in data representation, allowing for a combination of structured and unstructured data elements.

Importance of Structured Data

Plays a Big Role

Efficient Storage and Retrieval

Structured data can be stored and retrieved efficiently using relational database management systems (RDBMS).

Ease of Analysis

The organized nature of structured data simplifies data analysis and reporting tasks.

Data Integrity

Structured data typically adheres to predefined data integrity constraints, ensuring data consistency and accuracy.

Example:

In e-commerce, structured data containing product details such as SKU numbers, prices, and inventory levels facilitate seamless inventory management and order processing.

The Cornerstone of AI Algorithms: Structured Data

Structured Data is Important for AI Models

Structured data is the foundation upon which many powerful AI algorithms are built.

- **Clear and Consistent**

Structured data is well-organized and follows a defined format. This clarity allows AI algorithms to easily understand the data points and their relationships, enabling them to learn and make predictions with greater accuracy.

- **Efficient Processing**

The structured nature of the data facilitates faster processing by AI algorithms. Imagine searching for a specific book in a library; a well-organized catalog makes it much easier than sifting through a pile of unsorted books. Similarly, structured data allows AI algorithms to quickly access and analyze relevant information.

- **Reliable Training**

High-quality structured data is vital for training AI algorithms effectively. When an algorithm is trained on clean and consistent data, it develops a stronger understanding of the underlying patterns and can perform tasks with greater accuracy and reliability.

Scenario

How to Create a Dataset for Churn Analysis

Beginning of Month Snapshots of Active Customers

customer	month	product	tenure	tickets_30d	late_on_bill	churned
78658	Jan	phone	9	0	0	0
93343	Jan	internet & phone	4	2	0	0
54241	Jan	video & internet	15	0	0	1
76227	Jan	internet	9	0	0	0
77751	Jan	video & internet	9	0	1	1
5337	Jan	video & internet	47	0	0	0
37661	Jan	internet	21	1	0	0
55129	Jan	video & internet	14	0	0	0
78658	Feb	phone	10	0	0	0
93343	Feb	internet & phone	5	1	0	0
76227	Feb	internet	10	0	0	0
5337	Feb	video & internet	48	0	0	0
37661	Feb	internet	22	0	0	0
55129	Feb	video & internet	15	0	0	1
57496	Feb	phone	14	1	0	0
89127	Feb	video & internet	6	0	0	0
42563	Feb	internet	19	0	0	0

Churned Customers

customer	month
54241	Jan
77751	Jan
55129	Feb

This flag is created by using a left join on both customer ID and month

Source: zyabkina.com

Challenges and Opportunities of Unstructured Data

Things to Consider

Complexity

Unstructured data presents challenges in terms of organization, analysis, and interpretation due to its varied formats and content.

Scalability

Managing and processing large volumes of unstructured data can strain computational resources and storage systems.

Data Privacy and Security

Unstructured data may contain sensitive information, requiring robust security measures to protect privacy.

Challenges and Opportunities of Unstructured Data

Things To Consider

Rich Insights

Unstructured data holds valuable insights and patterns that can inform business decisions and drive innovation.

Advanced Analytics

Techniques such as text mining, sentiment analysis, and image recognition enable the extraction of meaningful insights from unstructured data.

Competitive Advantage

Effectively harnessing unstructured data can provide organizations with a competitive edge by uncovering hidden opportunities and trends.

Introduction To Artificial Intelligence

SUMMARY

- ✓ Structured data is well-organized and efficient for storage, retrieval, and analysis.
- ✓ Unstructured data lacks a predefined structure but holds valuable insights, requiring advanced techniques for analysis.
- ✓ Semi-structured data offers flexibility with a combination of structured and unstructured elements.
- ✓ Understanding these types of data is crucial for effective data management and analysis in AI and ML applications.



2.3

Data Collection

Lesson Learning Outcomes

- ✓ Understand the definition and significance of data collection and preprocessing in AI and ML
- ✓ Recognize the challenges posed by inaccurate, missing, imbalanced, and biased data
- ✓ Identify factors to consider before embarking on data collection, such as problem definition, model requirements, and available resources
- ✓ Explore common methods and approaches for collecting data, including leveraging internal sources, utilizing public datasets, web scraping, surveys, questionnaires, and APIs
- ✓ Gain insights into the importance of data quality criteria, including accuracy, completeness, consistency, and relevance, in ensuring reliable ML predictions

LESSON OVERVIEW

Module 2 Data Fundamentals

Lesson 2.3 Data Collection

In this lesson, we'll explore the crucial steps of data collection and preprocessing in the realm of **Artificial Intelligence (AI)** and **Machine Learning (ML)**. We'll start by defining data collection and preprocessing techniques, emphasizing their importance in ensuring the quality and reliability of data analysis and ML models. Next, we'll discuss the challenges associated with data, including inaccuracies, missing values, imbalances, and biases. Finally, we'll delve into practical considerations and methods for effective data collection, highlighting various approaches and common issues encountered.

- Explore crucial steps of data collection and preprocessing in AI and ML
- Define data collection and preprocessing techniques
- Delve into practical considerations and methods for effective data collection
- Highlight various approaches and common issues encountered

Data Collection

First Step for Analytics

Data Collection involves gathering raw data from various sources, while preprocessing techniques involve **cleaning, transforming**, and **preparing data** for analysis.

Importance

Effective data collection and preprocessing are critical for ensuring the quality and reliability of **Data Analysis** and **Machine Learning** models.

Challenges

- Inaccurate
- Missing
- Imbalanced and biased data

Setting the Stage: Defining your Data Needs



Before diving into data collection, it's essential to define your specific needs. What problem are you trying to solve, or what prediction are you aiming to make? Having a clear understanding of your goals helps determine the type and amount of data you need to collect. Consider factors like:

The problem you're addressing

What information is necessary to understand the problem and develop a solution?

The AI/ML model you plan to use

Different models have varying data requirements

The resources available

Data collection can be time-consuming and expensive. Determine the resources you have available for this stage

Exploring Data Collection Methods

Different Sources of Data

Various methods can be employed to collect data, depending on your needs and resources. Here are some common approaches:

Internal Data Sources

Leverage existing data within your organization, such as customer databases, transaction records, or sensor data.

Public Datasets

Numerous publicly available datasets can be a valuable resource for various domains (be mindful of licensing and usage restrictions).

Web Scraping

Extract data from websites using automated tools (ensure compliance with website terms of service and avoid copyright infringement).

Exploring Data Collection Methods

Different Sources of Data

Surveys and questionnaires

Collect data directly from users through surveys or questionnaires to gather specific information.

Application Programming Interfaces (APIs)

Access data from external sources through APIs provided by other organizations (requires proper authentication and authorization).

Importance of High-Quality Data

Key for Effective Algorithm

Training Basis

ML predictions rely on the quality of training data.

Common Issues

- Inaccurate Data
- Missing Data
- Data Imbalance
- Data Bias

Importance of High-Quality Data

Key for Effective Algorithm

Accuracy

The data should be free from errors and typos.

Completeness

All necessary data points should be present and accounted for.

Consistency

The data should follow a consistent format and use standard units of measurement.

Relevance

The data should be relevant to the problem you're trying to solve.

Data Collection

SUMMARY

- ✓ Data collection involves gathering raw data from various sources, while preprocessing techniques encompass cleaning, transforming, and preparing data for analysis.
- ✓ Effective data collection and preprocessing are crucial for ensuring the quality and reliability of data analysis and ML models.
- ✓ Challenges such as inaccuracies, missing values, imbalances, and biases must be addressed during data collection and preprocessing.
- ✓ Before collecting data, it's essential to define specific needs, considering factors like problem definition, model requirements, and available resources.
- ✓ Common methods for data collection include leveraging internal sources, utilizing public datasets, web scraping, surveys, questionnaires, and APIs.
- ✓ The quality of training data significantly impacts ML predictions, necessitating criteria such as accuracy, completeness, consistency, and relevance to ensure reliable outcomes.



2.4

Exploratory Data Analysis

Lesson Learning Outcomes

- ✓ Understand the definition and purpose of Exploratory Data Analysis (EDA) in data analysis and machine learning
- ✓ Recognize the importance of EDA in understanding data structure, relationships, and trends before applying advanced statistical techniques
- ✓ Learn the benefits of EDA, including uncovering hidden patterns, guiding feature selection, identifying data issues, and formulating hypotheses
- ✓ Explore various EDA techniques, including summary statistics, visualization, data profiling, and correlation analysis
- ✓ Gain practical skills in applying EDA techniques to analyze datasets effectively and prepare them for further analysis or modeling

LESSON OVERVIEW

Module 2 Data Fundamentals

Lesson 2.4 Exploratory Data Analysis

This lesson delves into **Exploratory Data Analysis (EDA)**, an essential approach for analyzing datasets to uncover their main characteristics using visual methods.

We'll explore the purpose and importance of EDA, aiming to gain insights, identify patterns, detect anomalies, and formulate hypotheses for further investigation before applying complex statistical techniques or building predictive models.

- Understand purpose and importance of EDA
- Gain insights, identify patterns, detect anomalies
- Formulate hypotheses before complex statistical techniques or model building

Exploratory Data Analysis (EDA)

The Way to Get Insights from Data

Exploratory Data Analysis (EDA) is an approach to analyzing datasets to summarize their main characteristics, often using visual methods.

Purpose

To gain insights into the data, identify patterns, detect anomalies, and formulate hypotheses for further investigation.

Importance

EDA helps researchers and analysts understand the structure, relationships, and underlying trends in the data before applying more complex statistical techniques or building predictive models.

Why is EDA Important?

Understanding Data is The Key Behind Building Effective Models

EDA offers several crucial benefits for data analysis and machine learning:

Uncover Hidden Gems

EDA helps us identify patterns, trends, and relationships within the data that might not be readily apparent from raw numbers.

Guide Feature Selection

By understanding the data's characteristics, we can select the most relevant features for building ML models, improving their accuracy and efficiency.

Identify Data Issues

EDA can reveal potential problems with the data, such as missing values, outliers, or inconsistencies, allowing us to address them before using the data for further analysis.

Formulate Hypotheses

Insights from EDA can help us formulate initial hypotheses about the data and guide further research or modeling efforts.

Techniques for EDA

Keep on Exploring !

Summary Statistics

Calculate basic statistics like mean, median, standard deviation, and frequency distributions to get a sense of central tendency and spread within the data.

Visualization

Create charts and graphs like histograms, scatter plots, boxplots, and heatmaps to visualize data distributions, identify outliers, and explore relationships between features.

Data Profiling

Analyze the characteristics of different variables in the data set, including data types, missing value rates, and unique value counts.

Correlation Analysis

Calculate correlation coefficients to measure the strength and direction of relationships between numerical features.

Exploratory Data Analysis

SUMMARY

- ✓ Exploratory Data Analysis (EDA) is crucial for understanding dataset characteristics before applying advanced statistical techniques or building predictive models.
- ✓ EDA helps uncover hidden patterns, guide feature selection, identify data issues, and formulate hypotheses for further investigation.
- ✓ Techniques such as summary statistics, visualization, data profiling, and correlation analysis are used in EDA to gain insights into data structure, relationships, and trends.
- ✓ Through EDA, researchers and analysts can gain a deeper understanding of the data, leading to more informed decision-making and improved model performance in machine learning applications.



2.5

Data Preprocessing

Lesson Learning Outcomes

- ✓ Understand the challenges associated with real-world data and the importance of preprocessing for ML readiness
- ✓ Explore techniques for handling missing values, outlier detection, deduplication, and data formatting to enhance data quality
- ✓ Learn approaches to handle outliers and transform data formats and scales for improved usability in ML models
- ✓ Gain insights into feature engineering techniques for creating informative features and addressing data imbalance
- ✓ Recognize the role of data visualization in aiding data preprocessing and understanding data distributions and relationships

LESSON OVERVIEW

Module 2 Data Fundamentals

Lesson 2.5 Data Preprocessing

This lesson introduces **Machine Learning (ML)**, a vital aspect of AI and computer science. We'll explore how ML mimics human learning using data and algorithms, emphasizing its iterative training process.

Inspired by human learning, ML's reliance on data drives its accuracy and efficiency, with applications spanning finance to healthcare, shaping our digital landscape.

- Explore critical steps of data processing and transformation for AI and ML.
- Discuss challenges like incompleteness, inconsistency, and noise in real-world data.
- Learn strategies for handling missing values, outliers, deduplication, and data formatting.
- Examine techniques for addressing outliers, transforming data, performing feature engineering, and managing data imbalance, along with the role of data visualization in aiding preprocessing.

Data Preprocessing

Enhancing Raw Data

- Real-world data is often incomplete, inconsistent, and noisy
- Preprocessing ensures data suitability for ML algorithms

Handling Missing Values

Imputation techniques such as mean, median, or mode replacement.

Outlier Detection and Removal

Identifying and removing data points that deviate significantly from the rest of the dataset.

Deduplication

Removing duplicate records to ensure data integrity.

Data Formatting

Converting data into a consistent format (e.g., date conversion).

Dealing with Outliers

Outliers are data points that fall significantly outside the expected range. Here are some approaches to handling outliers:

Investigate the Cause

Outliers can sometimes indicate errors or unexpected events. Investigate the cause before deciding on a correction method.

Capping

Set a threshold and replace outliers with the threshold value.

Removal

In some cases, removing outliers might be appropriate, especially if they are confirmed errors or irrelevant to your analysis.

Data Transformation

Data transformation involves modifying the format or scaling the values of your data to improve its usability for AI and ML models. Here are some common transformations:

Normalization

Scaling numerical data to a specific range (e.g., 0 to 1) to ensure all features have a similar influence on the model.

Standardization

Scaling numerical data to have a mean of 0 and a standard deviation of 1, often used for algorithms sensitive to feature scales.

Encoding Categorical Features

Converting categorical data (e.g., text labels) into numerical values that algorithms can understand (e.g., one-hot encoding).

Feature Engineering

Feature Engineering involves creating new features from existing data or selecting the most relevant features for your model. This step can significantly impact model performance. Here are some considerations:

Domain Knowledge

Leverage your understanding of the problem domain to identify potential features from the data.

Feature Selection

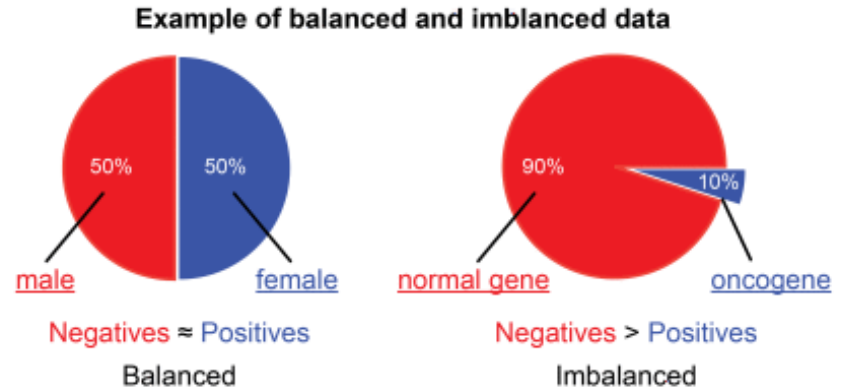
Choose features that are most informative and contribute significantly to the model's ability to learn.

Feature Creation

Derive new features by combining existing ones or applying mathematical transformations.

Balancing the Scales: Addressing Data Imbalance

- Data imbalance occurs when a dataset has a significant difference in the number of data points belonging to different classes
- This can lead to problems for AI and ML models, causing them to:
 - Be biased towards the majority class and perform poorly on the minority class
 - Not generalize well to unseen data with a more balanced class distribution



Balancing the Scales: Addressing Data Imbalance

Techniques to Address Data Imbalance

Undersampling: Reduce the number of data points in the majority class to match the size of the minority class.

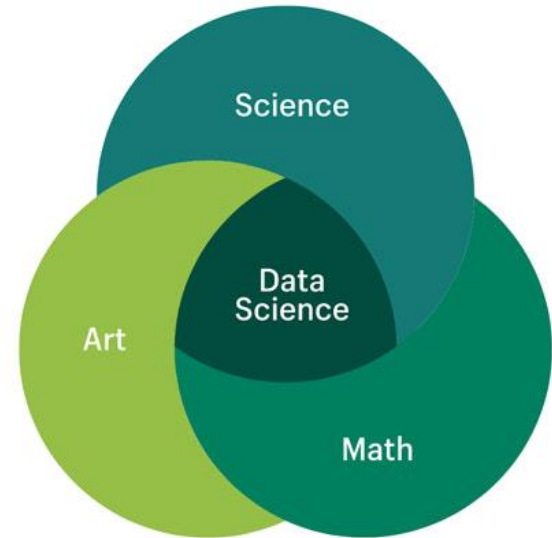
Example: Imagine you have 10,000 cat images and 100 snake images. Undersampling could involve randomly removing cat images until you have 100 of each class.

Oversampling: Replicate data points from the minority class to increase its representation in the dataset.

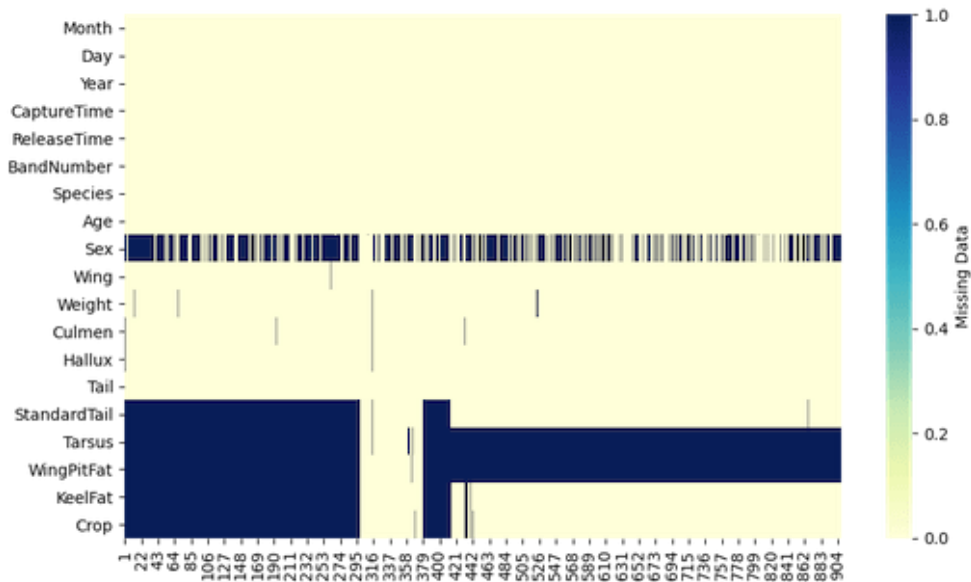
Example: Duplicate the 100 snake images multiple times to create a more balanced dataset with the same number of snake images as cat images.

Data Science Is A Mix of Art and Science !

Don't limit your thoughts only in existing theories



Data Visualization for Effective Preprocessing



Data visualization is a powerful tool that can aid in data preprocessing by helping us:

- Identify Missing Values and Outliers
- Understand Data Distribution
- Explore Relationships Between Features

Data Preprocessing

SUMMARY

- ✓ Data processing and transformation are essential steps for preparing raw data for analysis and modeling in AI and ML.
- ✓ Preprocessing techniques ensure data suitability by handling missing values, outliers, deduplication, and data formatting challenges.
- ✓ Techniques such as normalization, standardization, and encoding categorical features improve data usability for ML models.
- ✓ Feature engineering involves creating new features or selecting relevant ones to enhance model performance.
- ✓ Addressing data imbalance is crucial, and techniques like under-sampling and oversampling help balance class distributions.
- ✓ Data visualization aids in identifying missing values, and outliers, understanding data distributions, and exploring feature relationships during preprocessing.

Key Takeaways

We've embarked on a journey to understand the essential steps that lay the groundwork for successful AI and Machine Learning (ML) applications. Here's a summary of what we've learned:

Data Collection: We explored various methods for gathering data, emphasizing the importance of defining your specific needs and choosing the right approach based on resources and data availability.

Exploratory Data Analysis (EDA): We discovered the power of EDA as an initial investigation tool, uncovering hidden patterns, trends, and relationships within the data. We saw how EDA informs feature selection, identifies data issues, and helps us formulate initial hypotheses.

Data Preprocessing: We explored techniques for handling real-world data challenges and preprocessing for ML readiness, including handling missing values, outliers, and data formatting. Additionally, we learned about outlier handling, data transformation, feature engineering, and the role of data visualization in aiding data preprocessing for improved ML model usability.